Big Data Management
Wrap up

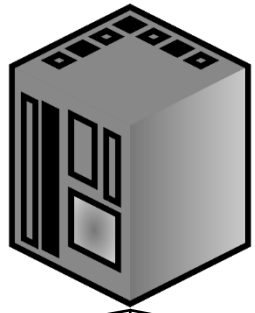
UCR
CELEBRATING 30 YEARS
Marlan and Rosemary Bourns
College of Engineering

What We Covered

- Storage (HDFS)
- Query processing (RDD, Hyracks)
- Higher-level data flow engines (SparkSQL, Structured Streaming)
- Storage formats (row, column, hybrid)
- Indexing (Global/local and LSM)
- Application-specific (Big Spatial Data and Machine Learning)
- End-to-end Big-data Management (AsterixDB)

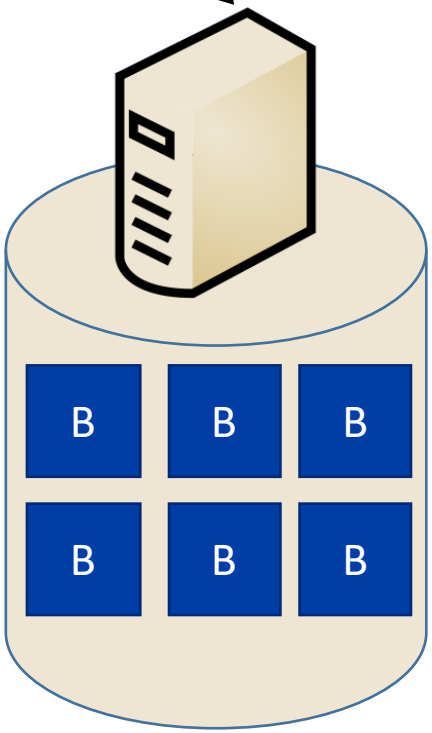
Big-data Storage (HDFS)

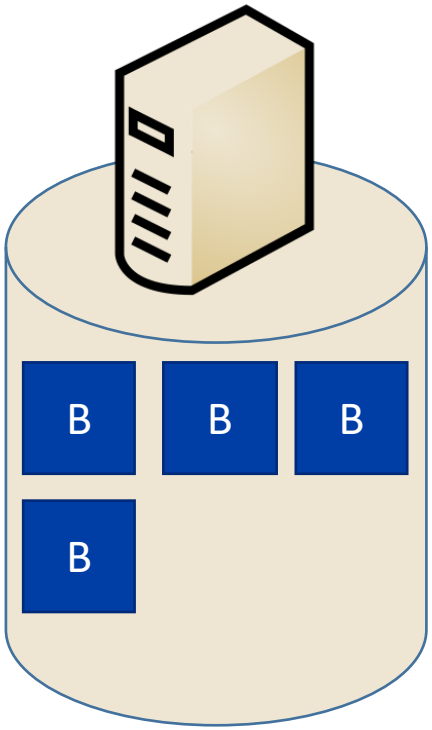
File and directory names
Block ordering and locations
Capacity of data nodes
Architecture of data nodes

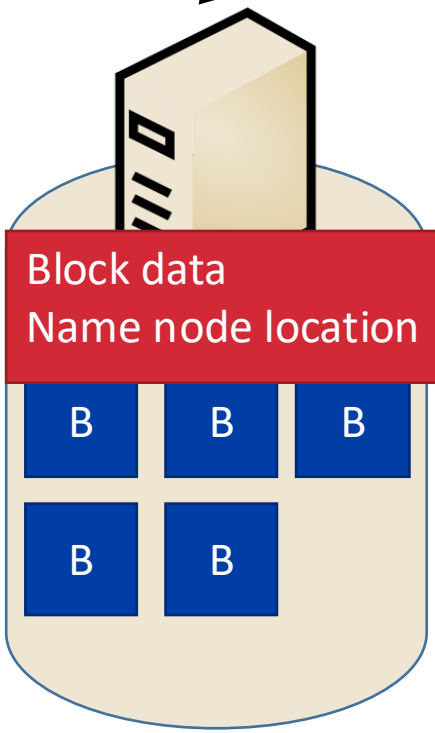


Name node

Data nodes

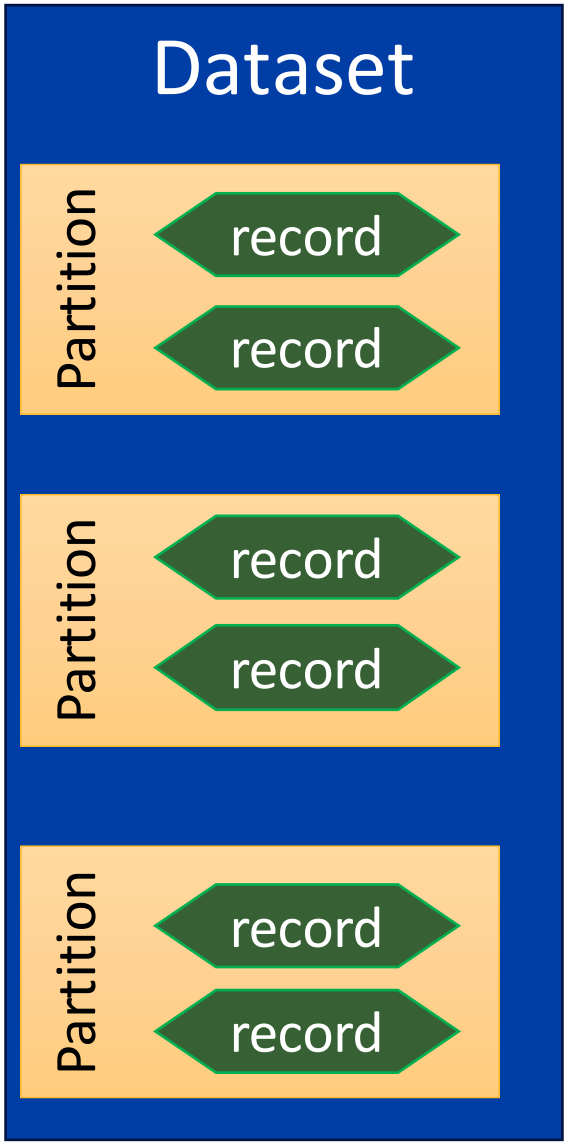




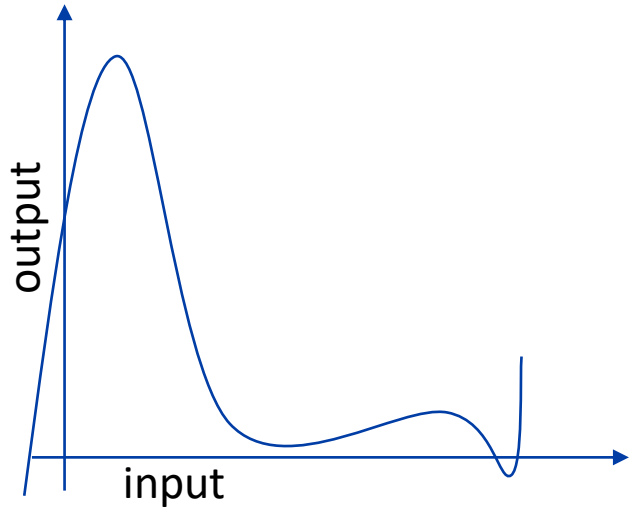


Block data
Name node location

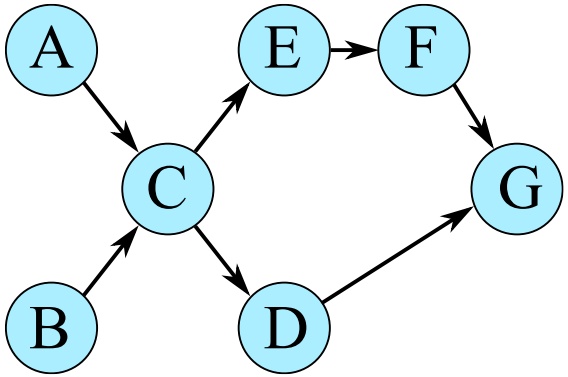
Big-data Processing



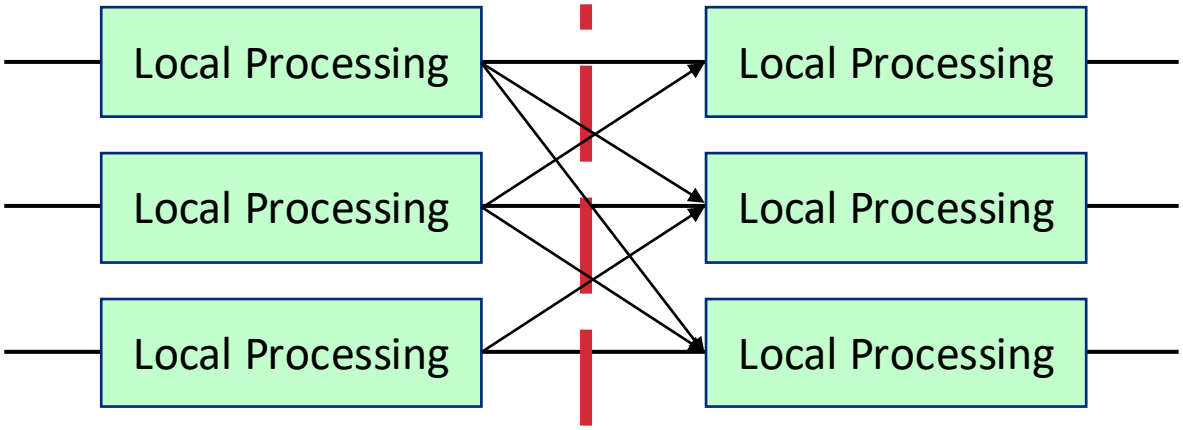
Data Model



Functional Programming Model



DAG Application Model



BSP Execution Model

Structured Data Processing

Dataframe

- Lazy execution
- Spark is aware of the data model
- Spark is aware of the query logic
- Can optimize the query

RDD

- Lazy execution
- The data model is hidden from Spark
- The transformations and actions are black boxes
- Cannot optimize the query

Dataframe = RDD + schema

Row/Column Storage Formats

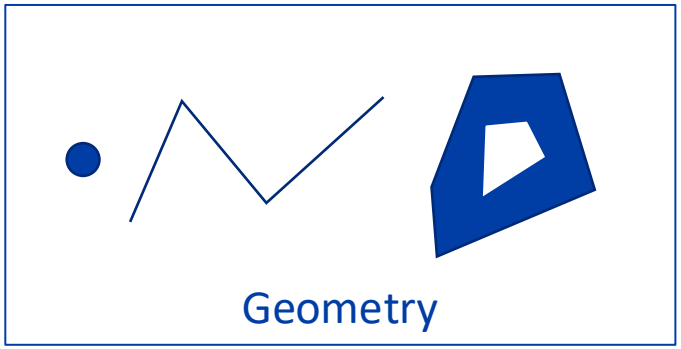
ID	Name	Email	…
1	Jack	jack@example.com	
2	Jill	jill@example.net	
3	Alex	alex@example.org	

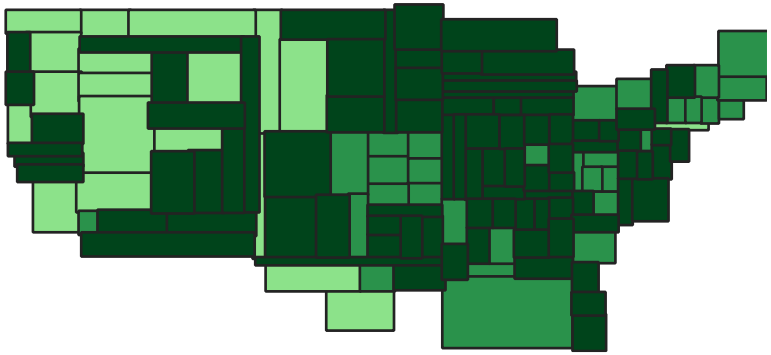
ID	Name	Email
1	Jack	…
2	Jill	…
3	Alex	…

Parquet

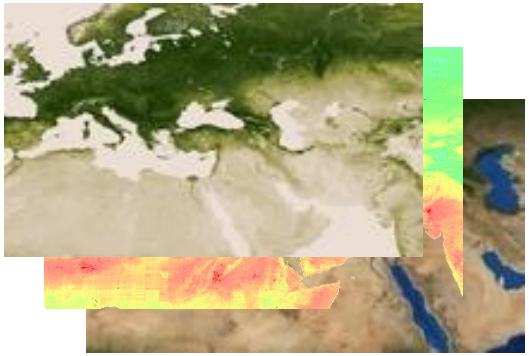
Big Spatial Data

Feature = Geometry + Other Attributes

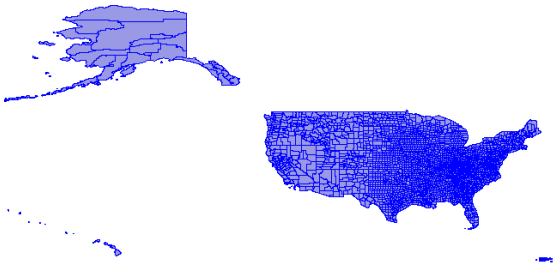


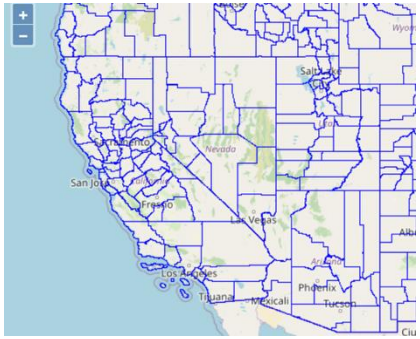


Spatial Partitioning



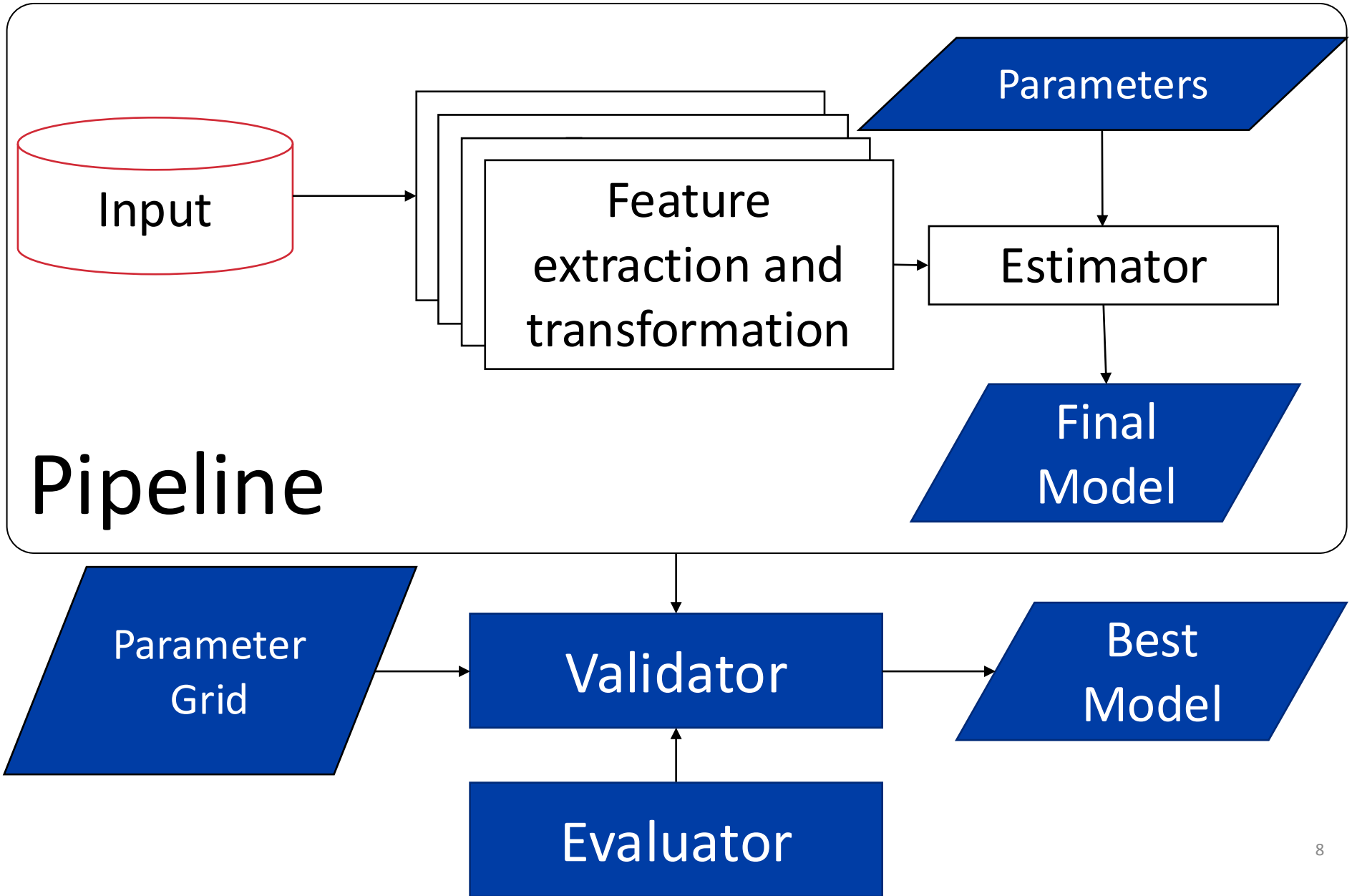
Raster Data Processing





Visualization

Machine Learning on Big-data



NoSQL and Document Databases

- Schema-less collections
  = Per-record schema
- Rich data types, nesting, repetition, geometry, text, …
- Rich indexes, multikey, text, location, …
- Conflicting data types for the same attribute

mongoDB®

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End-to-end Big-data Management

Semistructured
Data Management

Parallel
Database Systems

1st Generation
"Big Data" Systems

BDMS Desiderata:

- Able to **manage** data
- **Flexible** data model
- Full **query** capability
- Continuous data **ingestion**
- Efficient and robust **parallel** runtime
- Cost **proportional** to task at hand
- Support "**Big Data** data types"

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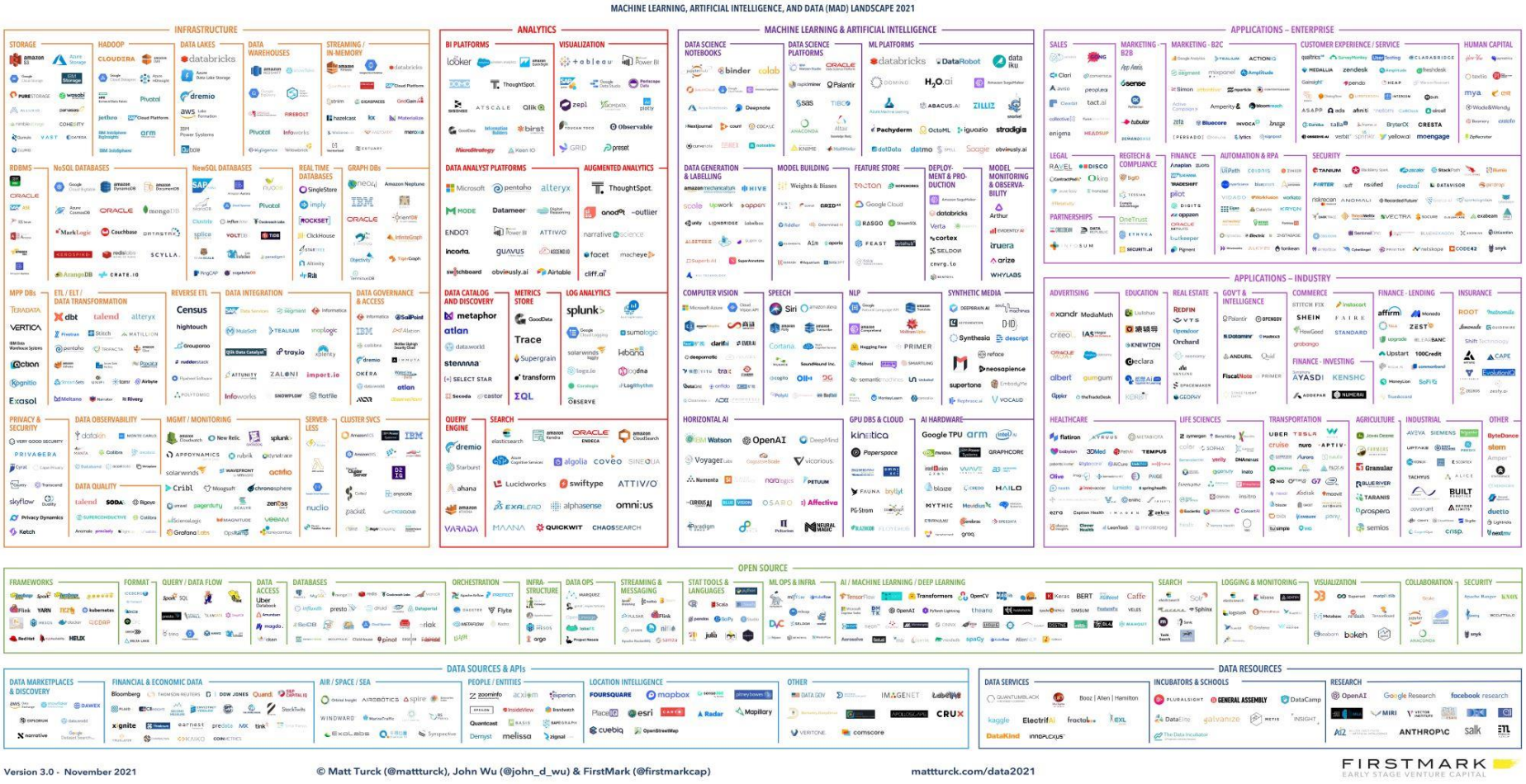
Machine Learning, AI, and Data (MAD) Landscape 2021



https://mattturck.com/data2021/

From Big Data Management to Big Data Science

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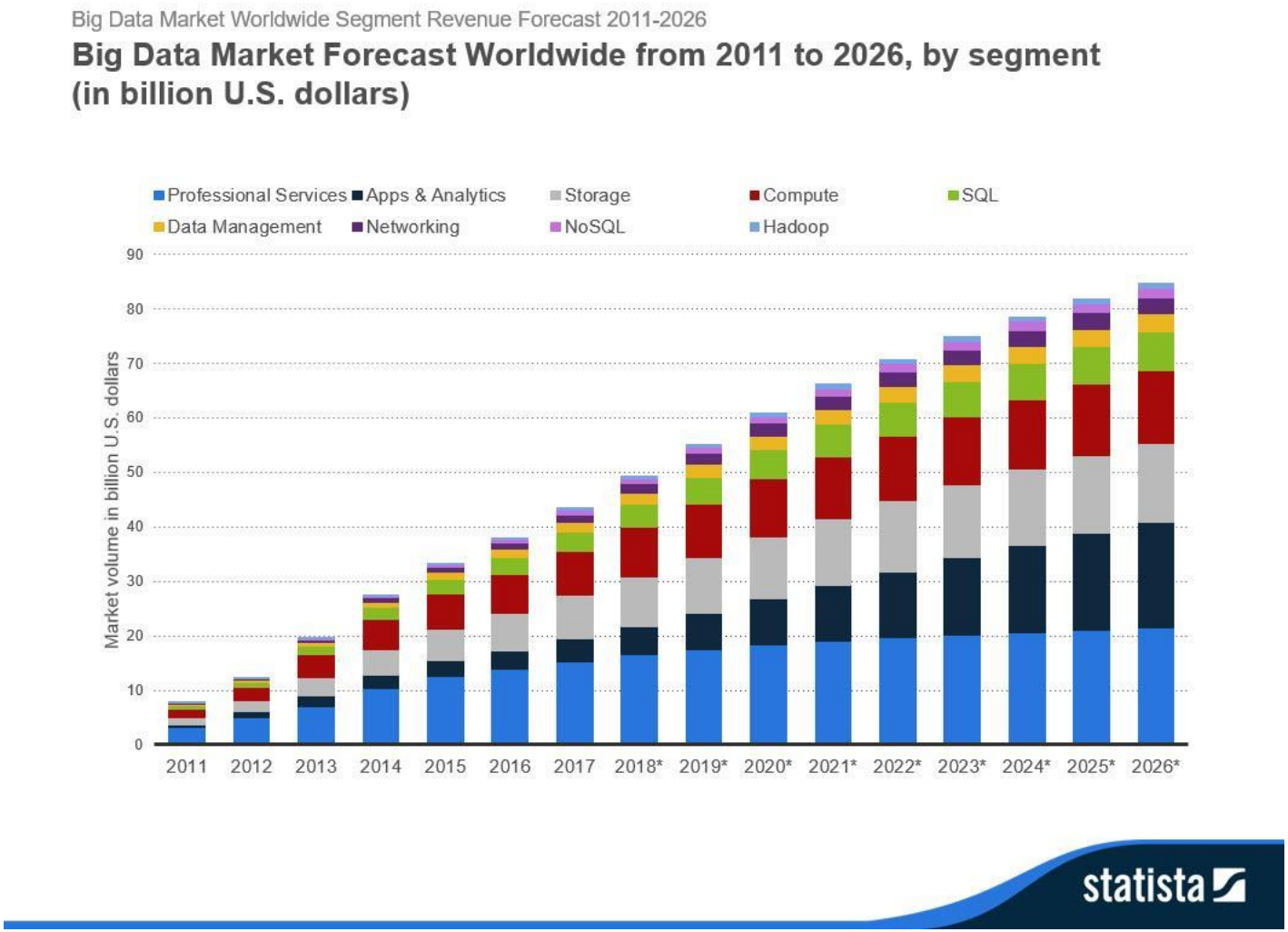
What is next?

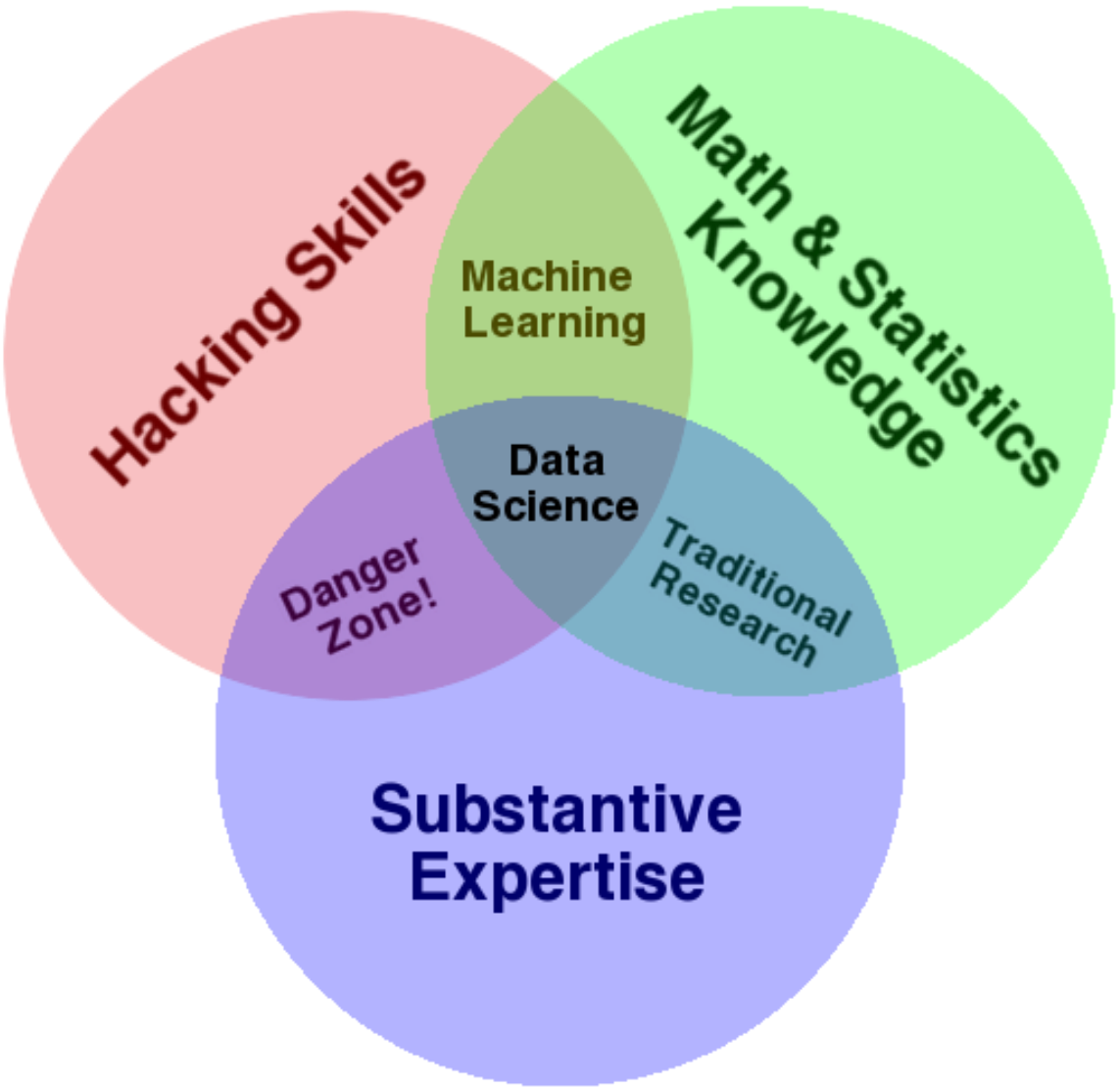
- Data is the new ~~oil~~ gold
- There is a shortage for experts in data management and analysis
- This course put you at the beginning of the track
- Keep your hands dirty and keep learning

Job Market



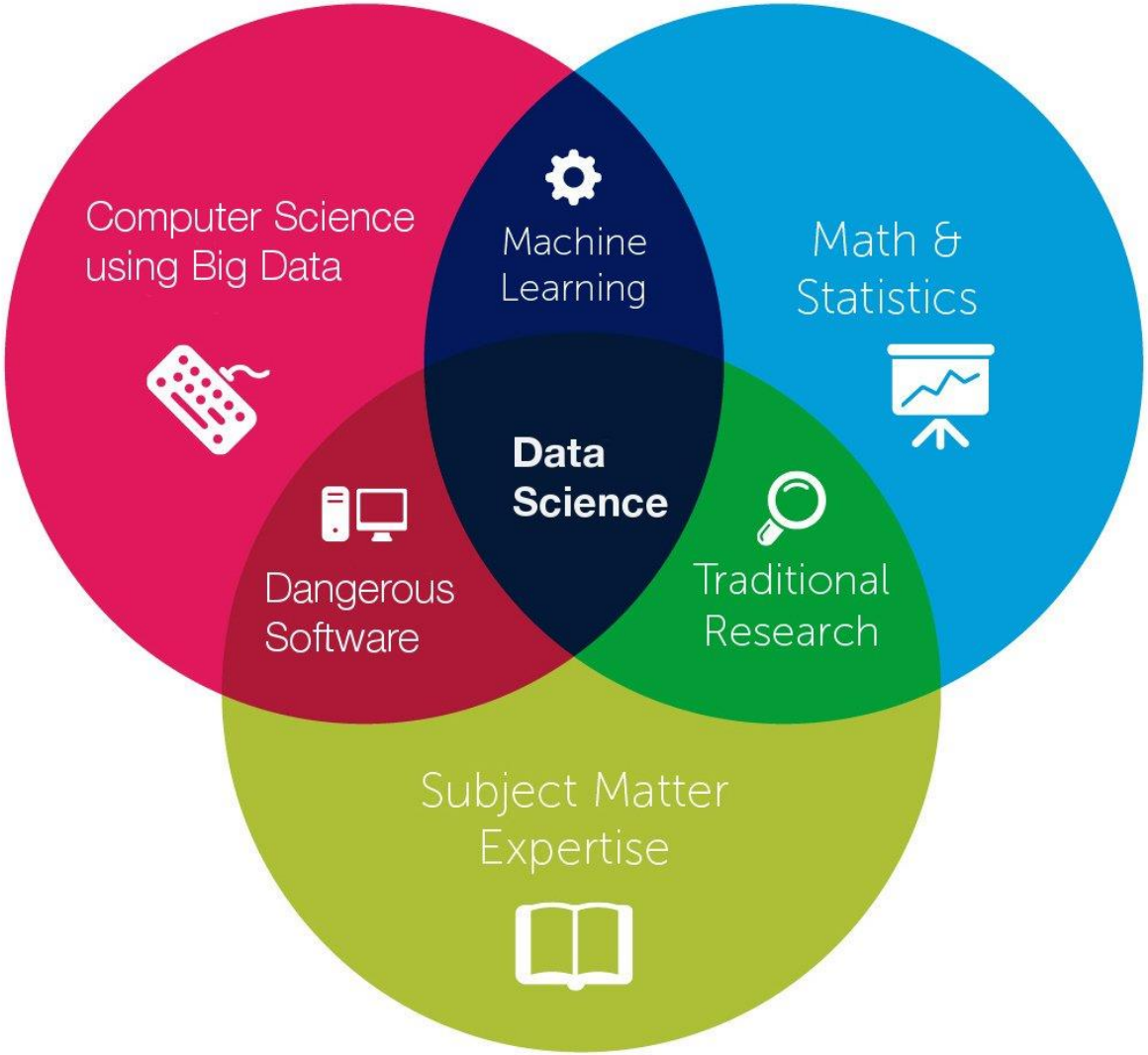
https://www.techicy.com/5-best-programming-languages-to-watch-out-in-2019-for-data-science.html

Data Science



Credits: Drew Conway

Data Science



https://mashimo.wordpress.com/2016/05/28/big-data-data-science-and-machine-learning-explained/

Data Scientist

Josh Wills
@josh_wills

Follow

Data Scientist (n.): Person who is better at statistics than any software engineer and better at software engineering than any statistician.

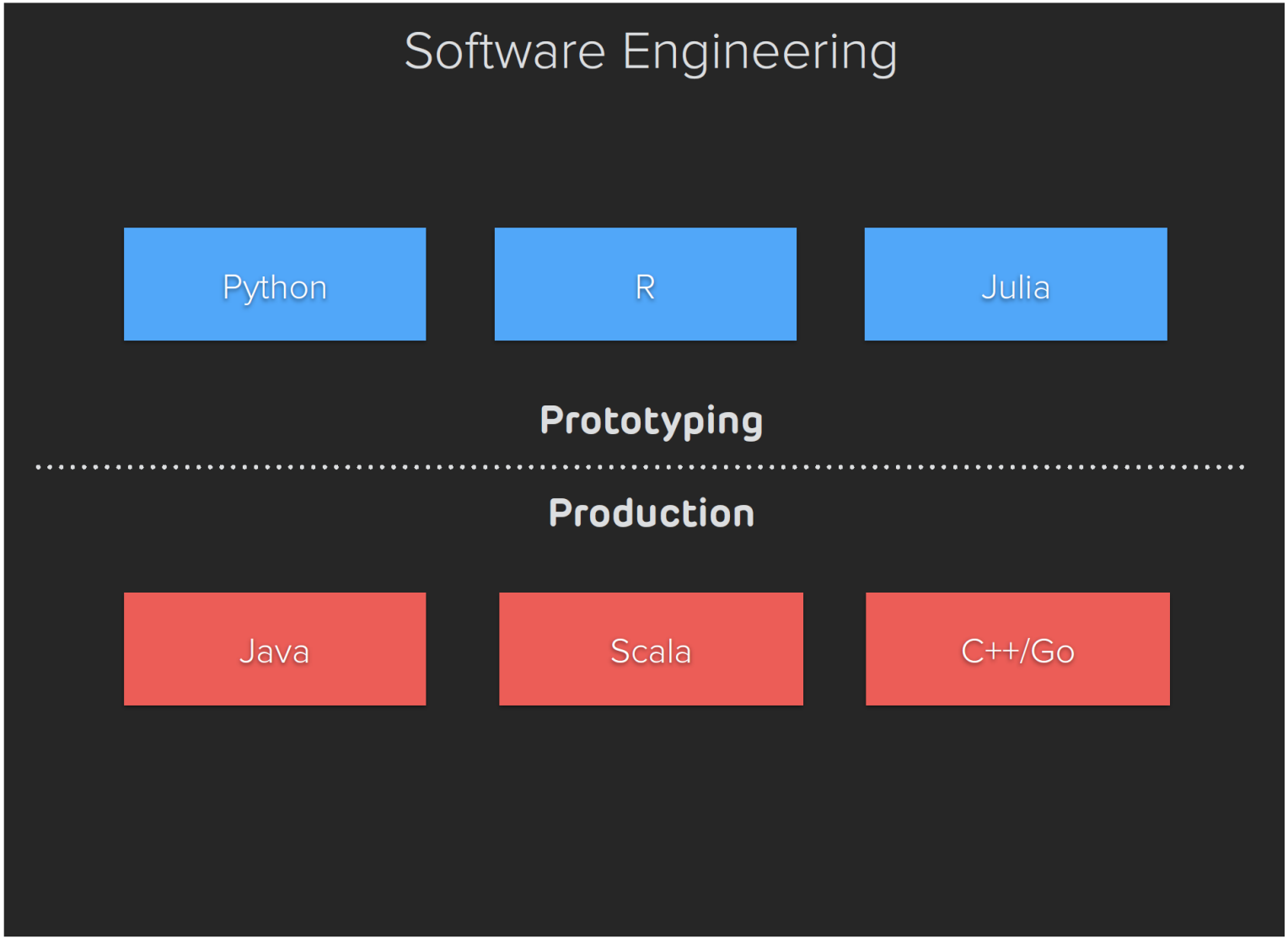
9:55 AM - 3 May 2012

1,725 Retweets **1,512** Likes

Next Steps

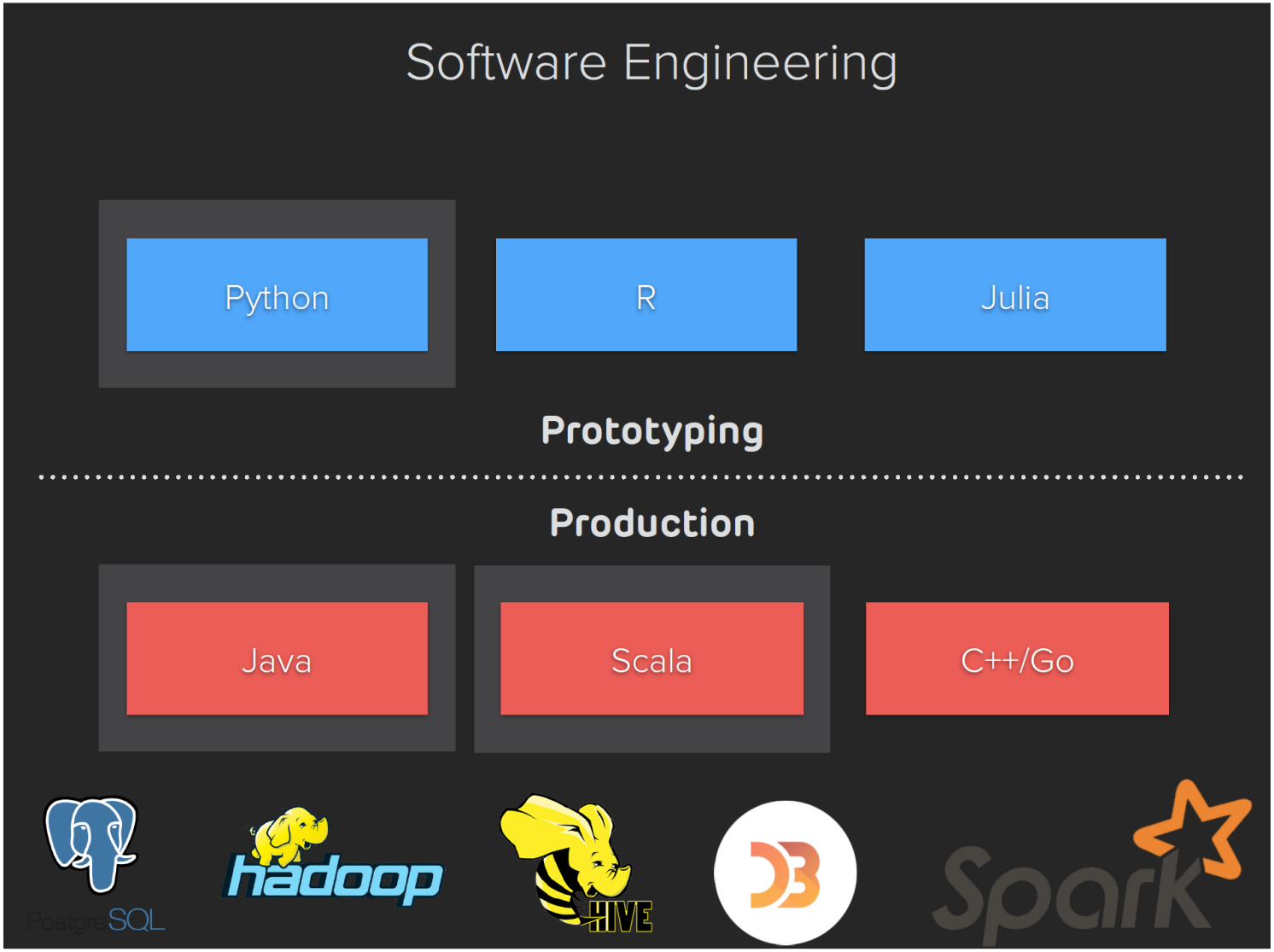
- CS
  - Big data tools
  - Python/R/Scala
- Math/Stats
  - Linear algebra
  - Correlation analysis
  - Hypothesis tests
- Collaboration with domain experts
  - Visualization
  - Prototyping

CS



https://www.slideshare.net/galvanizeHQ/how-to-become-a-data-scientist-by-ryan-orban-vp-of-operations-and-expansion-galvanize

CS/Big Data



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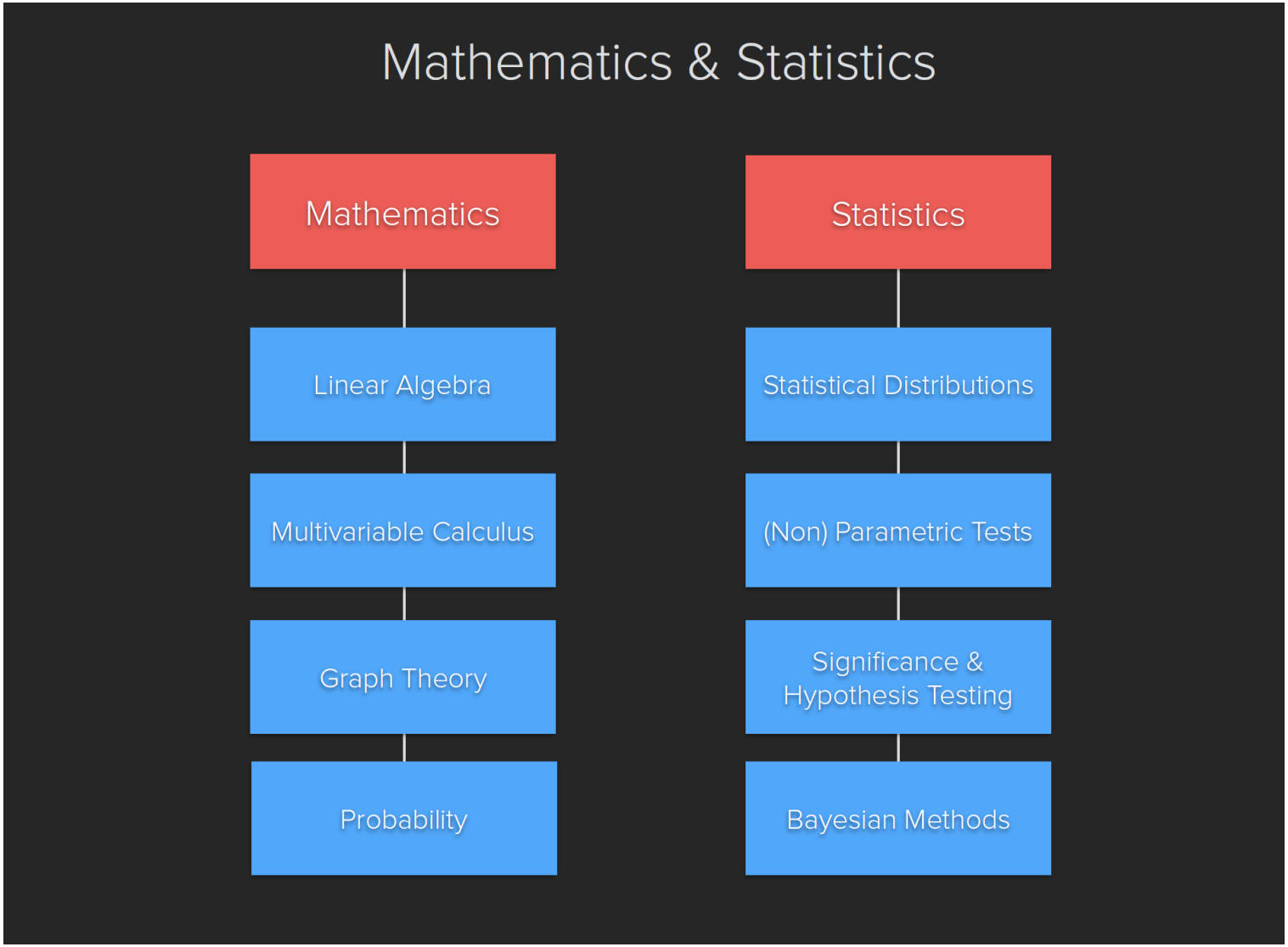
Math/Stats



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Online Courses

Mathematics & Statistics

galvanize

KHANACADEMY

Massachusetts Institute of Technology

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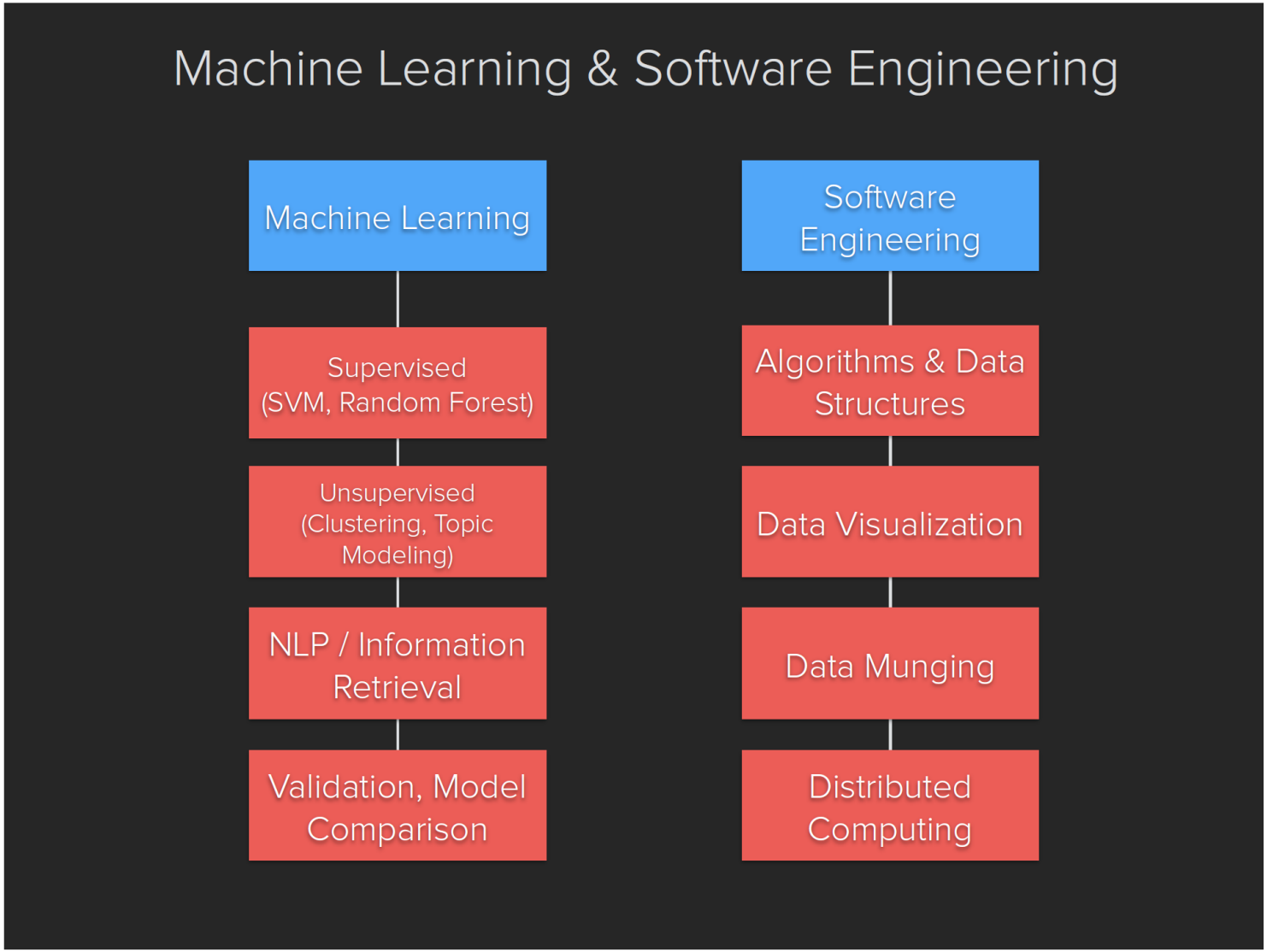
Data Analytics



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Big Data Landscape

Big data applications	MLlib	GraphX	Spatial	Streaming
NoSQL	Spark SQL	MongoDB	Algebricks/ SQL++	VDMBS
Query Processing	RDD	Hyracks	MapReduce	
Distributed Storage	HDFS	Parquet	LSM trees	

ASHES

Course Evaluation

- The course is quickly evolving
- Your anonymous feedback is highly valuable
- Both positive and negative comments are helpful
- Past students helped you. Please do your part and help future students.

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